DSAI541 Assignment 05

April 5, 2026

0.1 Assignment 5 - Deadline: April 5, 2026, Sun 11pm

DSAI 541 Spring 2026 Complete the assignment below and upload both the .ipynb file and its pdf to https://moodle.bogazici.edu.tr by the deadline given above. The submission page on Moodle will close automatically after this date and time and no late homeworks will be accepted.

To make a pdf, this may work if you're using Jupyter notebook on browser: Hit CMD+P or CTRL+P, and save it as PDF. You may also use other options from the File menu. (You can use Google Colab, but at the end you should download the ipynb notebook to submit it on Moodle.)

[1]:
```
# Run this cell first

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
"""import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers"""
from IPython.display import HTML
from sklearn.model_selection import train_test_split

# Set the display option to show all rows scrolling with a slider
pd.set_option('display.max_rows', None)
# To disable this, run the line below:
# pd.reset_option('display.max_rows')
```

You can use Tensorflow or PyTorch in this assignment.

1 Problem 1: MNIST with shuffled pixels (50 pts)

In class (Lecture 08.ipynb), we loaded MNIST data from Keras, normalized it, trained it and reached >96% accuracy. Your task is now:

Part a) For every image in the MNIST dataset, randomly shuffle its pixels with the same fixed shuffling function. To do this, generate a single random permutation of pixel positions and use it as a fixed shuffling function. Once defined, apply this function to all images in the dataset. Note: The shuffling function must remain the same for every image; all images should have their pixels rearranged using the exact same permutation. (If you shuffle each image with a random but different shuffling function, your dataset will be completely random and cannot be learned.)

Part b) Make a plot with MNISTplot2D(x_train[50]). The image should look like a noise, not as an image of a handwritten number since pixels are shuffled.

Part c) Train an ANN model as we did in Lecture 08.ipynb, but use early stopping. You should have gotten the same accuracy again around 96%. Explain why the model's accuracy did not drop significantly, even though we completely destroyed the spatial structure in the images where neighboring pixels were correlated due to image smoothness.

[2]:
```
# part a
```

[9]:
```
import torch
import torchvision
import torchvision.transforms as transforms
from torch.utils.data import DataLoader
from torch import nn, optim
import copy
```

[10]:
```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")
```

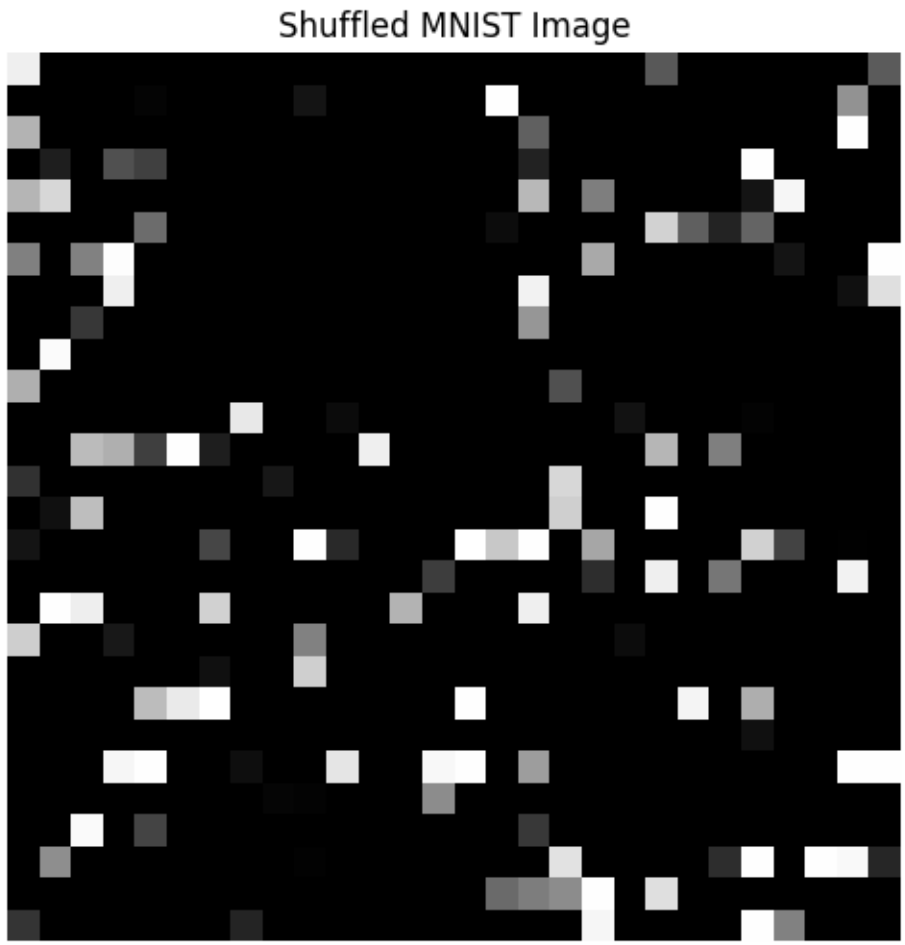
```
Using device: cuda
```

[ ]:
```
torch.manual_seed(3407) # set random seed for reproducibility
fixed_perm = torch.randperm(28*28) # fixed random permutation of pixel indices

def shuffling_transform(image: torch.Tensor, perm=fixed_perm):
    # Convert PIL Image to PyTorch tensor
    # Shuffle the pixels using the fixed permutation
    shuffled_image = image.view(-1)[perm].view(1, 28, 28)
    return shuffled_image

def inverse_shuffling_transform(image: torch.Tensor, perm=fixed_perm):
    # Inverse the shuffling by applying the inverse permutation
    inverse_perm = torch.argsort(perm)
    unshuffled_image = image.view(-1)[inverse_perm].view(1, 28, 28)
    return unshuffled_image


transform = transforms.Compose([
    transforms.ToTensor(), # Convert PIL Image to PyTorch tensor
    transforms.Normalize((0.1307,), (0.3081,)), # Normalize with mean and␣
 ↪standard deviation
    shuffling_transform # random shuffle the pixels in the image
])


# Load the MNIST dataset with the defined transform
```

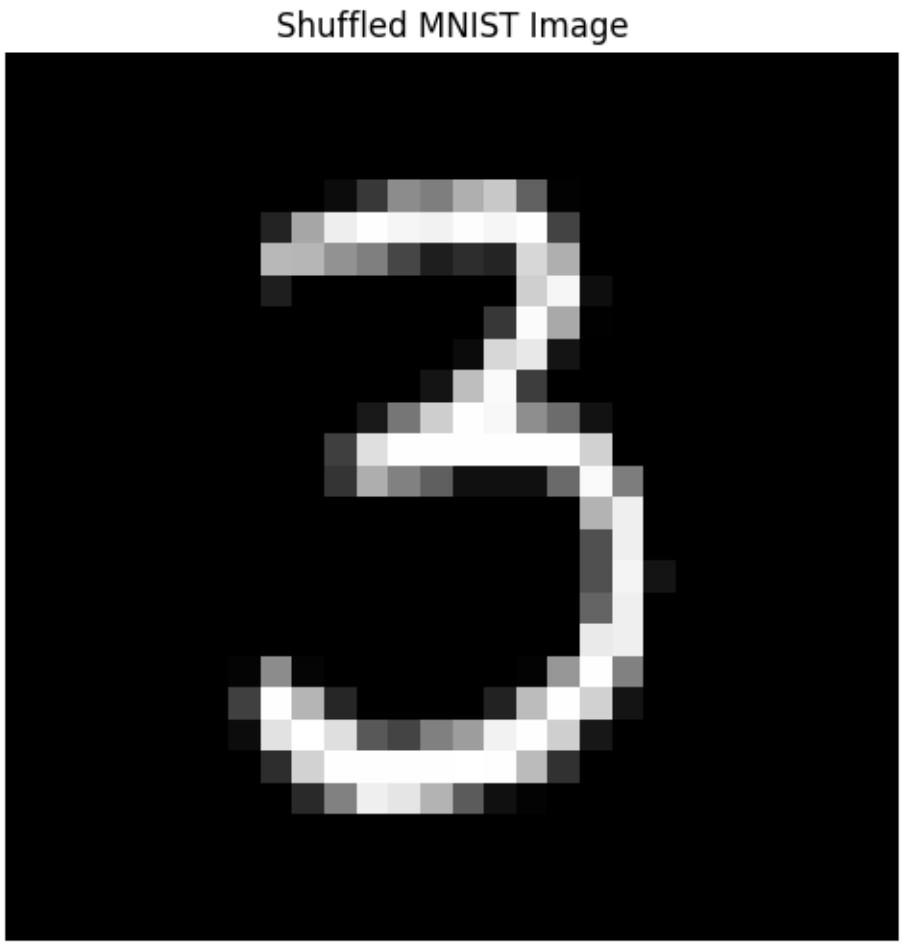
```
train_dataset = torchvision.datasets.MNIST(root='./data', train=True,␣
↪download=True, transform=transform)
test_dataset = torchvision.datasets.MNIST(root='./data', train=False,␣
↪download=True, transform=transform)
val_dataset, test_dataset = torch.utils.data.random_split(test_dataset, [5000,␣
↪5000], generator=torch.Generator().manual_seed(3407))

# Create data loaders for training, validation, and testing
train_loader = DataLoader(train_dataset, batch_size=64, shuffle=True)
val_loader = DataLoader(val_dataset, batch_size=64, shuffle=False)
test_loader = DataLoader(test_dataset, batch_size=64, shuffle=False)
```

[ ]:
```
# part b
```

[5]:
```
def MNISTplot2D(image):
    # Reshape the 1D tensor back to 28x28
    image_2d = image.view(28, 28)

    # Plot the image using seaborn heatmap
    plt.figure(figsize=(6, 6))
    sns.heatmap(image_2d.numpy(), cmap='gray', cbar=False, xticklabels=False,␣
↪yticklabels=False)
    plt.title('Shuffled MNIST Image')
    plt.show()

# Get a sample image from the training dataset
sample_image, sample_label = train_dataset[50]  # Get the first image and its␣
↪label
# Plot the sample image
MNISTplot2D(sample_image)
```



[6]:
```
# Original Image Before Shuffling

sample_image_unshuffled = inverse_shuffling_transform(sample_image)
MNISTplot2D(sample_image_unshuffled)
```



[ ]:
```
# part c
```

[ ]:
```
class MnistClassifier(nn.Module):
    def __init__(self):
        super(MnistClassifier, self).__init__()
        self.dense1 = nn.Linear(28*28, 25)  # Input layer to hidden layer
        self.dense2 = nn.Linear(25, 25)
        self.output = nn.Linear(25, 10)     # Hidden layer to output layer

    def forward(self, x):
        x = x.view(-1, 28*28)  # Flatten the input image
        x = torch.relu(self.dense1(x))  # First hidden layer with ReLU␣
↪activation
        x = torch.relu(self.dense2(x))  # Second hidden layer with ReLU␣
↪activation
        x = self.output(x)  # Output layer (logits)
        return x
```

[67]:
```
def train_model(model, train_loader, val_loader, optimizer, criterion,␣
↪scheduler=None, num_epochs=100):
    best_val_loss = float("inf")
    delta = 0.001
    best_model = copy.deepcopy(model.state_dict())
    last_improvement_epoch = 0

    train_losses = []
    val_losses = []

    for epoch in range(num_epochs):
        run_epoch_train_loss = 0
        model.train()
        for X_batch, y_batch in train_loader:
            X_batch = X_batch.to(device)
            y_batch = y_batch.to(device)
            optimizer.zero_grad()
            logits = model(X_batch)
            loss = criterion(logits , y_batch)
            loss.backward()
            optimizer.step()
            run_epoch_train_loss += loss.item()

        avg_train_loss = run_epoch_train_loss / len(train_loader)
        train_losses.append(avg_train_loss)

        print(f"Epoch {epoch+1}/{num_epochs}, Train Loss: {avg_train_loss:.
↪4f}", end='')

        # Validation phase
        model.eval()
        run_epoch_val_loss = 0
        with torch.no_grad():
            for X_batch, y_batch in val_loader:
                X_batch = X_batch.to(device)
                y_batch = y_batch.to(device) # Reshape to (batch_size, 1)
                logits = model(X_batch)
                loss = criterion(logits, y_batch)
                run_epoch_val_loss += loss.item()

        avg_val_loss = run_epoch_val_loss / len(val_loader)
        val_losses.append(avg_val_loss)

        print(f", Validation Loss: {avg_val_loss:.4f}")

        if scheduler is not None:
            scheduler.step(avg_val_loss)
```

```
        # Early stopping based on validation loss
        if avg_val_loss < best_val_loss - delta:
            best_val_loss = avg_val_loss
            best_model = copy.deepcopy(model.state_dict())
            last_improvement_epoch = epoch
        elif epoch - last_improvement_epoch >= 15: # Stop if no improvement for␣
↪15 epochs
            print(f"Early stopping at epoch {epoch} with best validation loss:␣
↪{best_val_loss:.4f}")
            break

    return train_losses, val_losses, best_val_loss, best_model,␣
↪last_improvement_epoch
```

[22]:
```
model = MnistClassifier().to(device)
criterion = nn.CrossEntropyLoss()
optimizer = optim.SGD(model.parameters(), lr=0.001, momentum=0.9, nesterov=True)
```

[23]:
```
train_losses, val_losses, best_val_loss, best_model, last_improvement_epoch =␣
↪train_model(model, train_loader, val_loader, optimizer, criterion,␣
↪num_epochs=100)
```

```
Epoch 1/100, Train Loss: 0.9466, Validation Loss: 0.3908
Epoch 2/100, Train Loss: 0.3565, Validation Loss: 0.2999
Epoch 3/100, Train Loss: 0.2961, Validation Loss: 0.2642
Epoch 4/100, Train Loss: 0.2605, Validation Loss: 0.2325
Epoch 5/100, Train Loss: 0.2343, Validation Loss: 0.2154
Epoch 6/100, Train Loss: 0.2135, Validation Loss: 0.1957
Epoch 7/100, Train Loss: 0.1962, Validation Loss: 0.1866
Epoch 8/100, Train Loss: 0.1827, Validation Loss: 0.1717
Epoch 9/100, Train Loss: 0.1711, Validation Loss: 0.1638
Epoch 10/100, Train Loss: 0.1613, Validation Loss: 0.1585
Epoch 11/100, Train Loss: 0.1525, Validation Loss: 0.1524
Epoch 12/100, Train Loss: 0.1454, Validation Loss: 0.1465
Epoch 13/100, Train Loss: 0.1396, Validation Loss: 0.1433
Epoch 14/100, Train Loss: 0.1333, Validation Loss: 0.1360
Epoch 15/100, Train Loss: 0.1279, Validation Loss: 0.1374
Epoch 16/100, Train Loss: 0.1231, Validation Loss: 0.1291
Epoch 17/100, Train Loss: 0.1189, Validation Loss: 0.1290
Epoch 18/100, Train Loss: 0.1149, Validation Loss: 0.1279
Epoch 19/100, Train Loss: 0.1111, Validation Loss: 0.1225
Epoch 20/100, Train Loss: 0.1076, Validation Loss: 0.1197
Epoch 21/100, Train Loss: 0.1051, Validation Loss: 0.1205
Epoch 22/100, Train Loss: 0.1021, Validation Loss: 0.1178
Epoch 23/100, Train Loss: 0.0998, Validation Loss: 0.1154
Epoch 24/100, Train Loss: 0.0970, Validation Loss: 0.1169
Epoch 25/100, Train Loss: 0.0948, Validation Loss: 0.1114
```

```
Epoch 26/100, Train Loss: 0.0927, Validation Loss: 0.1154
Epoch 27/100, Train Loss: 0.0907, Validation Loss: 0.1126
Epoch 28/100, Train Loss: 0.0884, Validation Loss: 0.1086
Epoch 29/100, Train Loss: 0.0868, Validation Loss: 0.1115
Epoch 30/100, Train Loss: 0.0851, Validation Loss: 0.1121
Epoch 31/100, Train Loss: 0.0835, Validation Loss: 0.1113
Epoch 32/100, Train Loss: 0.0816, Validation Loss: 0.1110
Epoch 33/100, Train Loss: 0.0801, Validation Loss: 0.1064
Epoch 34/100, Train Loss: 0.0785, Validation Loss: 0.1090
Epoch 35/100, Train Loss: 0.0768, Validation Loss: 0.1133
Epoch 36/100, Train Loss: 0.0758, Validation Loss: 0.1090
Epoch 37/100, Train Loss: 0.0740, Validation Loss: 0.1155
Epoch 38/100, Train Loss: 0.0731, Validation Loss: 0.1081
Epoch 39/100, Train Loss: 0.0714, Validation Loss: 0.1098
Epoch 40/100, Train Loss: 0.0700, Validation Loss: 0.1098
Epoch 41/100, Train Loss: 0.0693, Validation Loss: 0.1100
Epoch 42/100, Train Loss: 0.0674, Validation Loss: 0.1125
Epoch 43/100, Train Loss: 0.0668, Validation Loss: 0.1069
Early stopping at epoch 42 with best validation loss: 0.1064
```

[49]:
```python
# Plot training and validation losses
plt.figure(figsize=(10, 5))
plt.plot(train_losses, label='Train Loss')
plt.plot(val_losses, label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.title('Training and Validation Loss')
plt.legend()
plt.show()
```

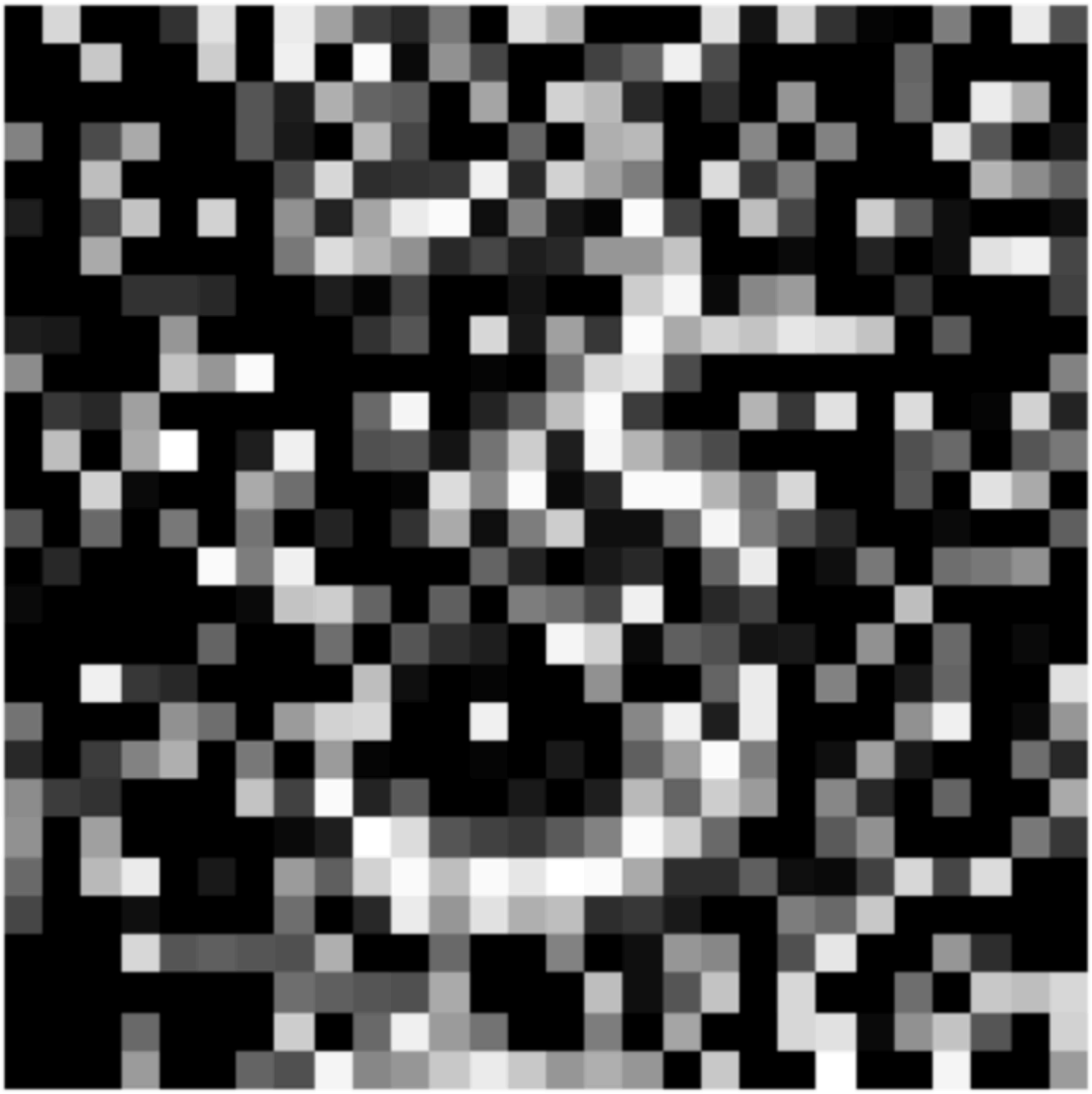
[25]:
```
model = MnistClassifier().to(device)
model.load_state_dict(best_model)
model.eval()


@torch.no_grad()
def evaluate_model(model, test_loader, criterion):
    test_loss = 0
    correct = 0
    total = 0

    for X_batch, y_batch in test_loader:
        X_batch = X_batch.to(device)
        y_batch = y_batch.to(device)
        logits = model(X_batch)
        loss = criterion(logits, y_batch)
        test_loss += loss.item()
        _, predicted = torch.max(logits.data, 1)
        total += y_batch.size(0)
        correct += (predicted == y_batch).sum().item()

    avg_test_loss = test_loss / len(test_loader)
    accuracy = correct / total
    return avg_test_loss, accuracy

test_loss, test_accuracy = evaluate_model(model, test_loader, criterion)
print(f"Test Loss: {test_loss:.4f}, Test Accuracy: {test_accuracy:.4f}")
```

```
Test Loss: 0.1174, Test Accuracy: 0.9652
```

2 Problem 2: Noisy MNIST (50 pts)

Part a) Write a function that takes an MNIST image, randomly selects 50% of its pixels, and assigns a random integer between 0 and 255 to each of these pixels. Each selected pixel should receive its own random value (though some pixels may end up with the same value by chance). Apply this function to all dataset to obtain noisy MNIST dataset. Note that this function will apply a different noise on each image as the function will be choosing different pixels and assigns random numbers to them each time it is called. Caution: Do this after loading the dataset `(x_train, y_train), (x_test, y_test) = mnist.load_data()` and before you normalize it.

Part b) Run `MNISTplot2D(x_train[50])`. It should output noisy "3" as below, where the handwritten number is still discernable by eye.



Part c) Train an ANN model as we did in Lecture 08.ipynb, but use early stopping. At what value did the accuracy fall? Explain why accuracy is lower in noisy MNIST when compared to the model trained in class on non-noisy, original MNIST data (its accuracy was 96%).

Part d) Now add ridge regression to the the dense layer of the model defined in Part c) to improve the model performance. You should be getting accuracy better than that in Part c). Explain why did ridge regression improved, if it did, the model.

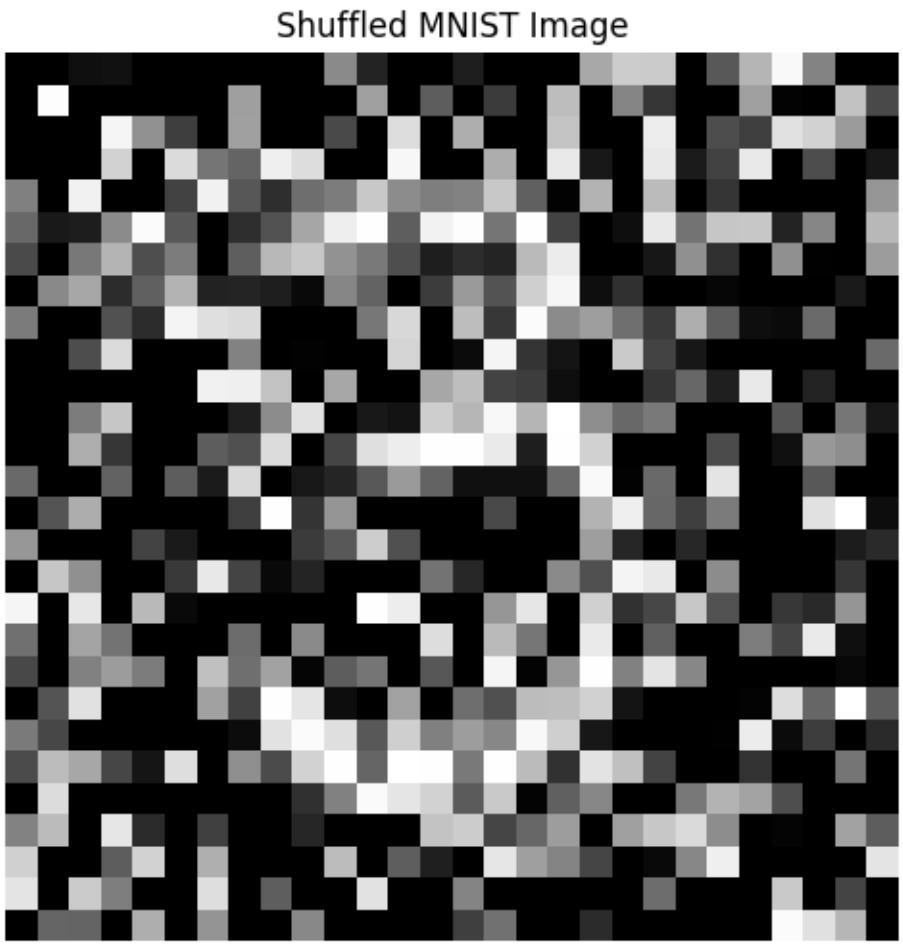
[ ]:
```
# part a
```

[45]:
```
def random_sel_replacement(image):
    # Flatten the image to a 1D tensor
    flat_image = image.view(-1)

    # Randomly select pixels to modify
    num_pixels = flat_image.size(0)
```

```
    num_to_modify = int(0.5 * num_pixels)
    indices_to_modify = torch.randperm(num_pixels)[:num_to_modify]
    modified_image = flat_image.clone()
    modified_image[indices_to_modify] = torch.randint(0, 256, (num_to_modify,)).
 ↪float() / 255.0  # Replace selected pixels with random values between 0 and␣
 ↪255 and normalize to [0, 1]
    return modified_image.view(1, 28, 28)  # Reshape back to original image␣
 ↪dimensions



transform = transforms.Compose([
    transforms.ToTensor(), # Convert PIL Image to PyTorch tensor
    random_sel_replacement, # random select pixels and replace them with random␣
 ↪values before normalization
    transforms.Normalize((0.1307,), (0.3081,)) # Normalize with mean and␣
 ↪standard deviation
])

train_dataset_2 = torchvision.datasets.MNIST(root='./data', train=True,␣
 ↪download=True, transform=transform)
test_dataset_2 = torchvision.datasets.MNIST(root='./data', train=False,␣
 ↪download=True, transform=transform)
val_dataset_2, test_dataset_2 = torch.utils.data.random_split(test_dataset_2,␣
 ↪[5000, 5000], generator=torch.Generator().manual_seed(3407))

# Create data loaders for training, validation, and testing
train_loader_2 = DataLoader(train_dataset_2, batch_size=64, shuffle=True)
val_loader_2 = DataLoader(val_dataset_2, batch_size=64, shuffle=False)
test_loader_2 = DataLoader(test_dataset_2, batch_size=64, shuffle=False)
```

[ ]:
```
# part b
```

[46]:
```
image = train_dataset_2[50][0]  # Get the first image from the original dataset␣
 ↪(before shuffling)
MNISTplot2D(image)
```



[ ]:
```
# part c
```

[68]:
```
model2 = MnistClassifier().to(device)
criterion2 = nn.CrossEntropyLoss()
optimizer2 = optim.SGD(model2.parameters(), lr=0.01, momentum=0.9,␣
↪nesterov=True)
scheduler = optim.lr_scheduler.ReduceLROnPlateau(optimizer2, patience=5,␣
↪factor=0.5)  # Reduce learning rate on plateau
```

[69]:
```
train_losses, val_losses, best_val_loss, best_model, last_improvement_epoch =␣
↪train_model(model2, train_loader_2, val_loader_2, optimizer2,␣
↪criterion2,scheduler, num_epochs=1000)
```

```
Epoch 1/1000, Train Loss: 0.7988, Validation Loss: 0.5382
Epoch 2/1000, Train Loss: 0.5189, Validation Loss: 0.4616
Epoch 3/1000, Train Loss: 0.4531, Validation Loss: 0.4023
Epoch 4/1000, Train Loss: 0.4206, Validation Loss: 0.3968
Epoch 5/1000, Train Loss: 0.3976, Validation Loss: 0.3488
```

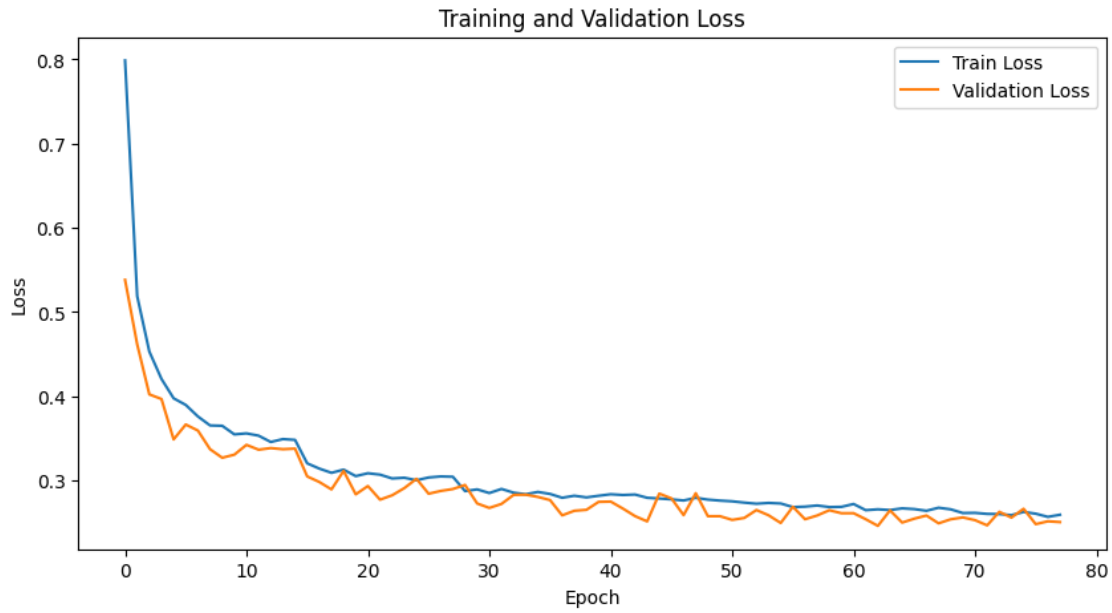
```
Epoch 6/1000, Train Loss: 0.3897, Validation Loss: 0.3665
Epoch 7/1000, Train Loss: 0.3760, Validation Loss: 0.3593
Epoch 8/1000, Train Loss: 0.3654, Validation Loss: 0.3372
Epoch 9/1000, Train Loss: 0.3650, Validation Loss: 0.3270
Epoch 10/1000, Train Loss: 0.3548, Validation Loss: 0.3307
Epoch 11/1000, Train Loss: 0.3560, Validation Loss: 0.3423
Epoch 12/1000, Train Loss: 0.3532, Validation Loss: 0.3366
Epoch 13/1000, Train Loss: 0.3457, Validation Loss: 0.3385
Epoch 14/1000, Train Loss: 0.3492, Validation Loss: 0.3371
Epoch 15/1000, Train Loss: 0.3482, Validation Loss: 0.3378
Epoch 16/1000, Train Loss: 0.3204, Validation Loss: 0.3049
Epoch 17/1000, Train Loss: 0.3141, Validation Loss: 0.2982
Epoch 18/1000, Train Loss: 0.3091, Validation Loss: 0.2894
Epoch 19/1000, Train Loss: 0.3129, Validation Loss: 0.3113
Epoch 20/1000, Train Loss: 0.3052, Validation Loss: 0.2836
Epoch 21/1000, Train Loss: 0.3086, Validation Loss: 0.2934
Epoch 22/1000, Train Loss: 0.3070, Validation Loss: 0.2773
Epoch 23/1000, Train Loss: 0.3024, Validation Loss: 0.2825
Epoch 24/1000, Train Loss: 0.3033, Validation Loss: 0.2910
Epoch 25/1000, Train Loss: 0.3003, Validation Loss: 0.3020
Epoch 26/1000, Train Loss: 0.3036, Validation Loss: 0.2843
Epoch 27/1000, Train Loss: 0.3048, Validation Loss: 0.2876
Epoch 28/1000, Train Loss: 0.3044, Validation Loss: 0.2899
Epoch 29/1000, Train Loss: 0.2874, Validation Loss: 0.2947
Epoch 30/1000, Train Loss: 0.2895, Validation Loss: 0.2727
Epoch 31/1000, Train Loss: 0.2851, Validation Loss: 0.2674
Epoch 32/1000, Train Loss: 0.2900, Validation Loss: 0.2722
Epoch 33/1000, Train Loss: 0.2855, Validation Loss: 0.2829
Epoch 34/1000, Train Loss: 0.2836, Validation Loss: 0.2832
Epoch 35/1000, Train Loss: 0.2864, Validation Loss: 0.2805
Epoch 36/1000, Train Loss: 0.2840, Validation Loss: 0.2768
Epoch 37/1000, Train Loss: 0.2795, Validation Loss: 0.2586
Epoch 38/1000, Train Loss: 0.2819, Validation Loss: 0.2640
Epoch 39/1000, Train Loss: 0.2799, Validation Loss: 0.2652
Epoch 40/1000, Train Loss: 0.2819, Validation Loss: 0.2746
Epoch 41/1000, Train Loss: 0.2837, Validation Loss: 0.2749
Epoch 42/1000, Train Loss: 0.2828, Validation Loss: 0.2667
Epoch 43/1000, Train Loss: 0.2833, Validation Loss: 0.2577
Epoch 44/1000, Train Loss: 0.2795, Validation Loss: 0.2514
Epoch 45/1000, Train Loss: 0.2786, Validation Loss: 0.2844
Epoch 46/1000, Train Loss: 0.2776, Validation Loss: 0.2787
Epoch 47/1000, Train Loss: 0.2763, Validation Loss: 0.2589
Epoch 48/1000, Train Loss: 0.2795, Validation Loss: 0.2849
Epoch 49/1000, Train Loss: 0.2773, Validation Loss: 0.2576
Epoch 50/1000, Train Loss: 0.2761, Validation Loss: 0.2576
Epoch 51/1000, Train Loss: 0.2753, Validation Loss: 0.2533
Epoch 52/1000, Train Loss: 0.2737, Validation Loss: 0.2555
Epoch 53/1000, Train Loss: 0.2725, Validation Loss: 0.2649
```

```
Epoch 54/1000, Train Loss: 0.2734, Validation Loss: 0.2587
Epoch 55/1000, Train Loss: 0.2727, Validation Loss: 0.2496
Epoch 56/1000, Train Loss: 0.2685, Validation Loss: 0.2688
Epoch 57/1000, Train Loss: 0.2690, Validation Loss: 0.2540
Epoch 58/1000, Train Loss: 0.2703, Validation Loss: 0.2584
Epoch 59/1000, Train Loss: 0.2684, Validation Loss: 0.2647
Epoch 60/1000, Train Loss: 0.2687, Validation Loss: 0.2611
Epoch 61/1000, Train Loss: 0.2721, Validation Loss: 0.2612
Epoch 62/1000, Train Loss: 0.2648, Validation Loss: 0.2540
Epoch 63/1000, Train Loss: 0.2658, Validation Loss: 0.2461
Epoch 64/1000, Train Loss: 0.2649, Validation Loss: 0.2645
Epoch 65/1000, Train Loss: 0.2669, Validation Loss: 0.2500
Epoch 66/1000, Train Loss: 0.2660, Validation Loss: 0.2547
Epoch 67/1000, Train Loss: 0.2640, Validation Loss: 0.2584
Epoch 68/1000, Train Loss: 0.2677, Validation Loss: 0.2491
Epoch 69/1000, Train Loss: 0.2658, Validation Loss: 0.2539
Epoch 70/1000, Train Loss: 0.2614, Validation Loss: 0.2561
Epoch 71/1000, Train Loss: 0.2616, Validation Loss: 0.2529
Epoch 72/1000, Train Loss: 0.2603, Validation Loss: 0.2467
Epoch 73/1000, Train Loss: 0.2601, Validation Loss: 0.2628
Epoch 74/1000, Train Loss: 0.2589, Validation Loss: 0.2559
Epoch 75/1000, Train Loss: 0.2627, Validation Loss: 0.2664
Epoch 76/1000, Train Loss: 0.2606, Validation Loss: 0.2482
Epoch 77/1000, Train Loss: 0.2568, Validation Loss: 0.2515
Epoch 78/1000, Train Loss: 0.2594, Validation Loss: 0.2506
Early stopping at epoch 77 with best validation loss: 0.2461
```

[71]:
```python
# print final lr
final_lr = optimizer2.param_groups[0]['lr']
print(f"Final Learning Rate: {final_lr:.6f}")
```

```
Final Learning Rate: 0.000156
```

[70]:
```python
# Plot training and validation losses
plt.figure(figsize=(10, 5))
plt.plot(train_losses, label='Train Loss')
plt.plot(val_losses, label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.title('Training and Validation Loss')
plt.legend()
plt.show()
```



[73]:
```
model2_test = MnistClassifier().to(device)
model2_test.load_state_dict(best_model)
model2_test.eval()


@torch.no_grad()
def evaluate_model(model, test_loader, criterion):
    test_loss = 0
    correct = 0
    total = 0

    for X_batch, y_batch in test_loader:
        X_batch = X_batch.to(device)
        y_batch = y_batch.to(device)
        logits = model(X_batch)
        loss = criterion(logits, y_batch)
        test_loss += loss.item()
        _, predicted = torch.max(logits.data, 1)
        total += y_batch.size(0)
        correct += (predicted == y_batch).sum().item()

    avg_test_loss = test_loss / len(test_loader)
    accuracy = correct / total
    return avg_test_loss, accuracy

test_loss, test_accuracy = evaluate_model(model2_test, test_loader_2, criterion)
```

```
print(f"Test Loss: {test_loss:.4f}, Test Accuracy: {test_accuracy:.4f}")
```

```
Test Loss: 0.2791, Test Accuracy: 0.9058
```

[ ]:
```
# Part d
```

[74]:
```
model3 = MnistClassifier().to(device)
criterion2 = nn.CrossEntropyLoss()
optimizer2 = optim.SGD(model3.parameters(), lr=0.01, momentum=0.9,␣
↪nesterov=True, weight_decay=1e-4)  # Add L2 regularization
scheduler = optim.lr_scheduler.ReduceLROnPlateau(optimizer2, patience=5,␣
↪factor=0.5)  # Reduce learning rate on plateau
```

[75]:
```
train_losses_3, val_losses_3, best_val_loss_3, best_model_3,␣
↪last_improvement_epoch_3 = train_model(model3, train_loader_2, val_loader_2,␣
↪optimizer2, criterion2,scheduler, num_epochs=1000)
```

```
Epoch 1/1000, Train Loss: 0.8000, Validation Loss: 0.5469
Epoch 2/1000, Train Loss: 0.5282, Validation Loss: 0.4494
Epoch 3/1000, Train Loss: 0.4627, Validation Loss: 0.4004
Epoch 4/1000, Train Loss: 0.4269, Validation Loss: 0.3870
Epoch 5/1000, Train Loss: 0.4082, Validation Loss: 0.3563
Epoch 6/1000, Train Loss: 0.3922, Validation Loss: 0.3944
Epoch 7/1000, Train Loss: 0.3816, Validation Loss: 0.3655
Epoch 8/1000, Train Loss: 0.3770, Validation Loss: 0.3623
Epoch 9/1000, Train Loss: 0.3676, Validation Loss: 0.3737
Epoch 10/1000, Train Loss: 0.3575, Validation Loss: 0.3522
Epoch 11/1000, Train Loss: 0.3595, Validation Loss: 0.3498
Epoch 12/1000, Train Loss: 0.3533, Validation Loss: 0.3423
Epoch 13/1000, Train Loss: 0.3479, Validation Loss: 0.3198
Epoch 14/1000, Train Loss: 0.3443, Validation Loss: 0.3200
Epoch 15/1000, Train Loss: 0.3405, Validation Loss: 0.3147
Epoch 16/1000, Train Loss: 0.3423, Validation Loss: 0.3400
Epoch 17/1000, Train Loss: 0.3415, Validation Loss: 0.3359
Epoch 18/1000, Train Loss: 0.3393, Validation Loss: 0.3360
Epoch 19/1000, Train Loss: 0.3367, Validation Loss: 0.3115
Epoch 20/1000, Train Loss: 0.3378, Validation Loss: 0.3368
Epoch 21/1000, Train Loss: 0.3355, Validation Loss: 0.3245
Epoch 22/1000, Train Loss: 0.3353, Validation Loss: 0.3188
Epoch 23/1000, Train Loss: 0.3357, Validation Loss: 0.3193
Epoch 24/1000, Train Loss: 0.3341, Validation Loss: 0.3184
Epoch 25/1000, Train Loss: 0.3362, Validation Loss: 0.3063
Epoch 26/1000, Train Loss: 0.3351, Validation Loss: 0.3110
Epoch 27/1000, Train Loss: 0.3274, Validation Loss: 0.2986
Epoch 28/1000, Train Loss: 0.3275, Validation Loss: 0.3258
Epoch 29/1000, Train Loss: 0.3330, Validation Loss: 0.3057
Epoch 30/1000, Train Loss: 0.3252, Validation Loss: 0.3311
Epoch 31/1000, Train Loss: 0.3275, Validation Loss: 0.3133
```

```
Epoch 32/1000, Train Loss: 0.3256, Validation Loss: 0.3207
Epoch 33/1000, Train Loss: 0.3286, Validation Loss: 0.3215
Epoch 34/1000, Train Loss: 0.3078, Validation Loss: 0.2935
Epoch 35/1000, Train Loss: 0.2986, Validation Loss: 0.3137
Epoch 36/1000, Train Loss: 0.2965, Validation Loss: 0.2862
Epoch 37/1000, Train Loss: 0.2982, Validation Loss: 0.2868
Epoch 38/1000, Train Loss: 0.2981, Validation Loss: 0.2811
Epoch 39/1000, Train Loss: 0.2962, Validation Loss: 0.2807
Epoch 40/1000, Train Loss: 0.2944, Validation Loss: 0.2860
Epoch 41/1000, Train Loss: 0.2930, Validation Loss: 0.2971
Epoch 42/1000, Train Loss: 0.2920, Validation Loss: 0.2860
Epoch 43/1000, Train Loss: 0.2911, Validation Loss: 0.2833
Epoch 44/1000, Train Loss: 0.2906, Validation Loss: 0.2783
Epoch 45/1000, Train Loss: 0.2898, Validation Loss: 0.2732
Epoch 46/1000, Train Loss: 0.2927, Validation Loss: 0.2865
Epoch 47/1000, Train Loss: 0.2923, Validation Loss: 0.3063
Epoch 48/1000, Train Loss: 0.2923, Validation Loss: 0.2834
Epoch 49/1000, Train Loss: 0.2880, Validation Loss: 0.2966
Epoch 50/1000, Train Loss: 0.2894, Validation Loss: 0.2811
Epoch 51/1000, Train Loss: 0.2861, Validation Loss: 0.2983
Epoch 52/1000, Train Loss: 0.2810, Validation Loss: 0.2717
Epoch 53/1000, Train Loss: 0.2744, Validation Loss: 0.2680
Epoch 54/1000, Train Loss: 0.2781, Validation Loss: 0.2651
Epoch 55/1000, Train Loss: 0.2772, Validation Loss: 0.2810
Epoch 56/1000, Train Loss: 0.2732, Validation Loss: 0.2550
Epoch 57/1000, Train Loss: 0.2743, Validation Loss: 0.2679
Epoch 58/1000, Train Loss: 0.2743, Validation Loss: 0.2767
Epoch 59/1000, Train Loss: 0.2764, Validation Loss: 0.2691
Epoch 60/1000, Train Loss: 0.2685, Validation Loss: 0.2571
Epoch 61/1000, Train Loss: 0.2725, Validation Loss: 0.2610
Epoch 62/1000, Train Loss: 0.2708, Validation Loss: 0.2696
Epoch 63/1000, Train Loss: 0.2694, Validation Loss: 0.2715
Epoch 64/1000, Train Loss: 0.2663, Validation Loss: 0.2467
Epoch 65/1000, Train Loss: 0.2684, Validation Loss: 0.2652
Epoch 66/1000, Train Loss: 0.2638, Validation Loss: 0.2566
Epoch 67/1000, Train Loss: 0.2609, Validation Loss: 0.2596
Epoch 68/1000, Train Loss: 0.2644, Validation Loss: 0.2582
Epoch 69/1000, Train Loss: 0.2629, Validation Loss: 0.2618
Epoch 70/1000, Train Loss: 0.2596, Validation Loss: 0.2414
Epoch 71/1000, Train Loss: 0.2629, Validation Loss: 0.2452
Epoch 72/1000, Train Loss: 0.2626, Validation Loss: 0.2479
Epoch 73/1000, Train Loss: 0.2592, Validation Loss: 0.2460
Epoch 74/1000, Train Loss: 0.2637, Validation Loss: 0.2574
Epoch 75/1000, Train Loss: 0.2644, Validation Loss: 0.2471
Epoch 76/1000, Train Loss: 0.2633, Validation Loss: 0.2556
Epoch 77/1000, Train Loss: 0.2587, Validation Loss: 0.2477
Epoch 78/1000, Train Loss: 0.2575, Validation Loss: 0.2504
Epoch 79/1000, Train Loss: 0.2599, Validation Loss: 0.2435
```

```
Epoch 80/1000, Train Loss: 0.2537, Validation Loss: 0.2450
Epoch 81/1000, Train Loss: 0.2579, Validation Loss: 0.2485
Epoch 82/1000, Train Loss: 0.2549, Validation Loss: 0.2531
Epoch 83/1000, Train Loss: 0.2555, Validation Loss: 0.2536
Epoch 84/1000, Train Loss: 0.2551, Validation Loss: 0.2389
Epoch 85/1000, Train Loss: 0.2545, Validation Loss: 0.2517
Epoch 86/1000, Train Loss: 0.2560, Validation Loss: 0.2431
Epoch 87/1000, Train Loss: 0.2570, Validation Loss: 0.2517
Epoch 88/1000, Train Loss: 0.2555, Validation Loss: 0.2585
Epoch 89/1000, Train Loss: 0.2548, Validation Loss: 0.2569
Epoch 90/1000, Train Loss: 0.2572, Validation Loss: 0.2548
Epoch 91/1000, Train Loss: 0.2567, Validation Loss: 0.2404
Epoch 92/1000, Train Loss: 0.2541, Validation Loss: 0.2423
Epoch 93/1000, Train Loss: 0.2550, Validation Loss: 0.2453
Epoch 94/1000, Train Loss: 0.2518, Validation Loss: 0.2545
Epoch 95/1000, Train Loss: 0.2492, Validation Loss: 0.2317
Epoch 96/1000, Train Loss: 0.2563, Validation Loss: 0.2486
Epoch 97/1000, Train Loss: 0.2533, Validation Loss: 0.2419
Epoch 98/1000, Train Loss: 0.2577, Validation Loss: 0.2464
Epoch 99/1000, Train Loss: 0.2562, Validation Loss: 0.2350
Epoch 100/1000, Train Loss: 0.2558, Validation Loss: 0.2372
Epoch 101/1000, Train Loss: 0.2570, Validation Loss: 0.2513
Epoch 102/1000, Train Loss: 0.2547, Validation Loss: 0.2509
Epoch 103/1000, Train Loss: 0.2544, Validation Loss: 0.2519
Epoch 104/1000, Train Loss: 0.2557, Validation Loss: 0.2453
Epoch 105/1000, Train Loss: 0.2565, Validation Loss: 0.2496
Epoch 106/1000, Train Loss: 0.2545, Validation Loss: 0.2339
Epoch 107/1000, Train Loss: 0.2515, Validation Loss: 0.2454
Epoch 108/1000, Train Loss: 0.2513, Validation Loss: 0.2526
Epoch 109/1000, Train Loss: 0.2556, Validation Loss: 0.2481
Epoch 110/1000, Train Loss: 0.2568, Validation Loss: 0.2455
Early stopping at epoch 109 with best validation loss: 0.2317
```

[77]:
```python
# Plot training and validation losses
plt.figure(figsize=(10, 5))
plt.plot(train_losses_3, label='Train Loss')
plt.plot(val_losses_3, label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.title('Training and Validation Loss')
plt.legend()
plt.show()
```

[76]:
```
model3_test = MnistClassifier().to(device)
model3_test.load_state_dict(best_model_3)
model3_test.eval()


@torch.no_grad()
def evaluate_model(model, test_loader, criterion):
    test_loss = 0
    correct = 0
    total = 0

    for X_batch, y_batch in test_loader:
        X_batch = X_batch.to(device)
        y_batch = y_batch.to(device)
        logits = model(X_batch)
        loss = criterion(logits, y_batch)
        test_loss += loss.item()
        _, predicted = torch.max(logits.data, 1)
        total += y_batch.size(0)
        correct += (predicted == y_batch).sum().item()

    avg_test_loss = test_loss / len(test_loader)
    accuracy = correct / total
    return avg_test_loss, accuracy
```

```
test_loss, test_accuracy = evaluate_model(model3_test, test_loader_2,␣
↪criterion2)
print(f"Test Loss: {test_loss:.4f}, Test Accuracy: {test_accuracy:.4f}")
```

```
Test Loss: 0.2537, Test Accuracy: 0.9172
```